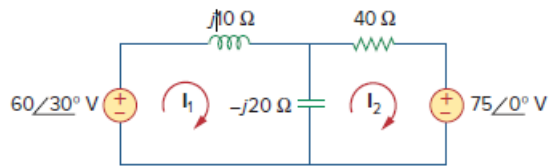


Problem

Using mesh analysis, find I_1 and I_2 in the circuit of Fig. 10.75.

Figure 10.75:



Step-by-step solution

Step 1 of 3

Refer to circuit diagram in Figure 10.75 in the textbook.

Apply Kirchhoff's voltage law to mesh 1.

$$j10I_1 + (I_1 - I_2)(-j20) - 60\angle 30^\circ = 0$$

$$(j10 - j20)I_1 + I_2(j20) = 60\angle 30^\circ$$

$$-(j10)I_1 + I_2(j20) = 60\angle 30^\circ$$

$$I_2(j2) = 6\angle 30^\circ + (j1)I_1$$

$$I_2 = \frac{6\angle 30^\circ + (j1)I_1}{j2}$$

$$I_2 = 3\angle -60^\circ + 0.5I_1 \dots\dots (1)$$

Step 2 of 3

Apply Kirchhoff's voltage law to mesh 2.

$$(I_2 - I_1)(-j20) + 40I_2 + 75\angle 0^\circ = 0$$

$$(j20)I_1 + (40 - j20)I_2 + 75\angle 0^\circ = 0$$

Substitute $3\angle -60^\circ + 0.5I_1$ for I_2 .

$$(j20)I_1 + (40 - j20)(3\angle -60^\circ + 0.5I_1) + 75\angle 0^\circ = 0$$

$$(j20 + 20 - j10)I_1 + 8.038 - j133.92 + 75 = 0$$

$$(20 + j10)I_1 = -83.038 + j133.92$$

$$I_1 = \frac{-83.038 + j133.92}{20 + j10}$$

$$= -0.643 + j7.017$$

$$= 7.047\angle 95.24^\circ \text{ A}$$

Therefore, the value of current, I_1 is $7.047\angle 95.24^\circ \text{ A}$.

Step 3 of 3

Recall equation (1).

$$\mathbf{I}_2 = 3\angle -60^\circ + 0.5\mathbf{I}_1$$

Substitute $7.047\angle 95.24^\circ \text{ A}$ for $\mathbf{I}_1$.

$$\begin{aligned}\mathbf{I}_2 &= 3\angle -60^\circ + (0.5)(7.047\angle 95.24^\circ) \\ &= 1.5 - j2.598 - 0.32 + j3.5088 \\ &= 1.18 + j0.91069 \\ &= 1.4892\angle 37.71^\circ \text{ A}\end{aligned}$$

Therefore, the value of current, $\mathbf{I}_2$ is $\boxed{1.4892\angle 37.71^\circ \text{ A}}$.